Hyper-Parameter Optimization Report

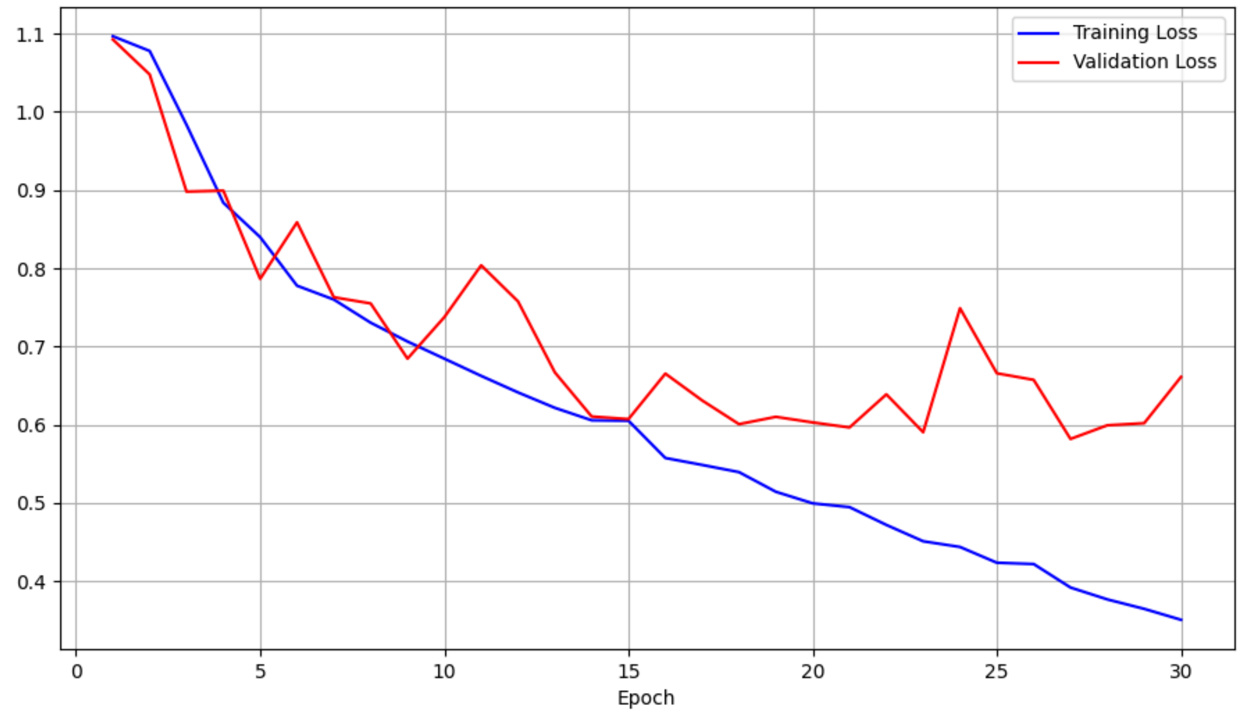
Optimized Network

Model: "sequential"

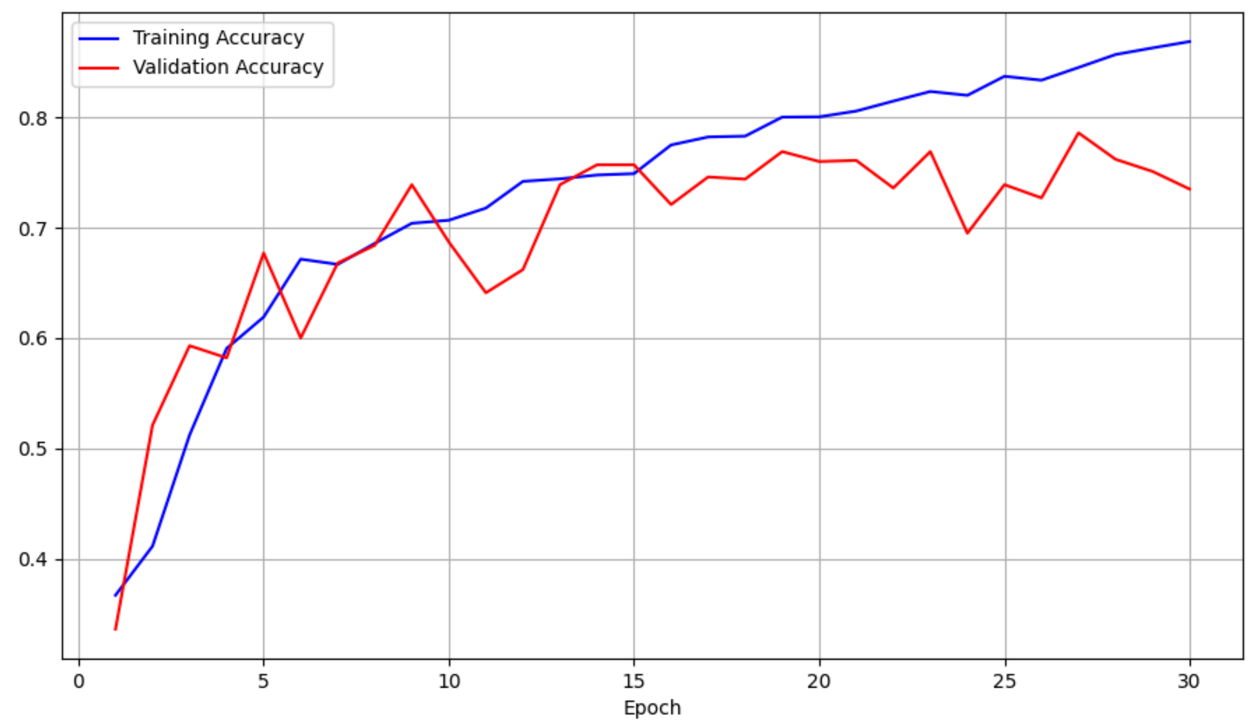
Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 147, 147, 16)	784
max_pooling2d (MaxPooling2D)	(None, 36, 36, 16)	0
dropout (Dropout)	(None, 36, 36, 16)	0
conv2d_1 (Conv2D)	(None, 33, 33, 32)	8224
max_pooling2d_1 (MaxPooling2D)	(None, 11, 11, 32)	0
dropout_1 (Dropout)	(None, 11, 11, 32)	0
conv2d_2 (Conv2D)	(None, 8, 8, 48)	24624
max_pooling2d_2 (MaxPooling 2D)	(None, 4, 4, 48)	0
dropout_2 (Dropout)	(None, 4, 4, 48)	0
flatten (Flatten)	(None, 768)	0
dense (Dense)	(None, 64)	49216
dropout_3 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 3)	195

Total params: 83,043
Trainable params: 83,043
Non-trainable params: 0

Plot Showing Training and Validation Loss as Function of Epoch:



Plot Showing Accuracy Against the Training and Validation Sets as a Function of Epoch:



Accuracy and Loss of Best Learned Model:
Epoch 30/30
32/32 [==============================] - 12s 374ms/step - loss: 0.3508 - accuracy: 0.8687 - val_loss: 0.6612 - val_accuracy: 0.7350

Test Set Evaluation:
* Evaluating basic_model
30/30 [==============================] - 3s 95ms/step - loss: 0.7129 - accuracy: 0.7082

Hyper-Parameter Optimization Strategy Report:

- Convolution layers
    - Old Version: 2 convolution layers
    - New Version: 3 convolution layers
    - Effect: This let our model learn more abstract features which helped us have a higher accuracy potential.
- Convolution Kernels
    - Old Version: Kernel size was set to default for Conv2D
    - New Version: Kernel size was set to (4,4) for all Conv2D
    - Effect: The larger kernel size helped increase computation efficiency as well as let the model learn broader and chunkier features.
- Pooling Windows
    - Old Version: (4,4) for both convolution layers
    - New Version: Changing pooling sizes (4,4), (3,3), (2,2)
    - Effect: The larger pooling windows at the start helped reduce the number of parameters for the later convolution layers which resulted in faster training.
- Dropout Strategy
    - Old Version: No dropout
    - New Version: We added Dropouts after each convolution block and dense layer
    - Effect: Adding the dropouts helped us have better generalization and reduce overfitting
- Filter Changes
    - Old Version: Two Filters 16, 32
    - New Version: Three Filters 16, 32, 48
    - Effect: Adding another filter (from adding the third convolution layer) increased the number of feature types learned which helped the model have a higher accuracy ceiling.
- Performance Impact
    - Old model achieved a 64.75% accuracy
    - New model achieved a 70.82% accuracy
    - Best validation accuracy increased from 60.10% to 73.50%